

Exercise 6

Consider N non-interacting spin- $\frac{1}{2}$ magnetic moments μ in an external magnetic field B . The two energy levels are

$$E_{\uparrow} = -\mu B, \quad E_{\downarrow} = +\mu B.$$

- Evaluate the final temperature for

$$T_i = 1.2 \text{ K}, \quad B_i = 2.0 \text{ T}, \quad B_f = 2.0 \times 10^{-2} \text{ T}.$$

- Find the limiting forms of the entropy:

$$S(B, T) \quad \text{for} \quad x \ll 1 \quad \text{and} \quad x \gg 1.$$

- Interpret these two limits physically.

Exercise 7

Suppose all fundamental interactions conserve parity exactly. We want to explain to aliens what "left" and "right" mean.

- Explain why the coordinate systems

$$(x, y, z) \quad \text{and} \quad (-x, y, z)$$

would be physically equivalent in such a world.

- Explain why parity-violating weak interactions remove this ambiguity.